



## Project Plan

Data Analytics Capstone

**Course Code:** MGMT 6219

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# **1. Problem Statement & Research Questions**

## **1.1 Business Challenge Summary**

Canada Vapes operates in a national market that is not truly uniform. Although the company sells to customers across Canada, buyer behavior may vary significantly by province because of differences in customer preferences, product availability, shipping expectations, provincial regulations, taxation, local competition, and market maturity (Canada Vapes, n.d.). Canada Vapes has been serving Canadian customers since 2010, providing a strong base of historical customer and transaction data that can be used to better understand where growth opportunities exist.

The core business challenge is that Canada Vapes needs to identify which Canadian provinces offer the strongest growth potential and determine how marketing strategies should be adapted by region. Treating Canada as a single market may lead to generic campaigns that do not reflect local buying patterns. For example, one province may show high average order value but low repeat purchase frequency, while another may have strong repeat purchasing but lower basket size. Similarly, provinces with stricter flavor, advertising, or tax rules may require more compliance-focused product positioning than provinces with fewer restrictions.

This project will use internal Canada Vapes sales and customer data, combined with external market and regulatory research, to compare provincial buyer behaviour and recommend where the company should focus growth efforts. The analysis will examine metrics such as revenue by province, average order value, purchase frequency, repeat purchase behavior, product category preferences, and possible customer lifetime value indicators. These insights will be compared against external factors such as vape excise duties, provincial restrictions, advertising limitations, and competitor presence.

This approach follows the analytics lifecycle, where a project begins with a clearly defined business problem before moving into data collection, cleaning, exploration, analysis, interpretation, and recommendations (Davenport & Harris, 2017). It also aligns with the importance of framing the right business question to ensure that data analysis supports meaningful decision-making (Provost & Fawcett, 2013).

A major consideration is that vaping is a highly regulated product category. Federal rules restrict vaping product advertising that can be seen or heard by young people, and point-of-sale promotion is also limited. Vaping advertisements generally require health warnings unless an exception applies (Government of Canada, 2020). In addition, federal excise duty applies to vaping products across Canada, and additional vaping duty

applies in specified provinces. As of 2026, specified vaping provinces include Alberta, Manitoba, New Brunswick, Northwest Territories, Nunavut, Ontario, Prince Edward Island, Quebec, Yukon, and Nova Scotia (Canada Revenue Agency, 2026). This means regional growth recommendations must be realistic, compliant, and focused on legal-age adult consumers rather than broad mass-market promotion.

Therefore, the purpose of this project is not simply to identify where Canada Vapes currently sells the most. The goal is to determine where the company has the best combination of current performance, customer value, regulatory feasibility, and future growth opportunities. The final recommendations should help Canada Vapes make informed decisions about which provinces to prioritize, what products to emphasize, and what marketing messages are most appropriate for each priority region.

## **1.2 Research Questions**

### **1. How does buyer behavior differ across Canadian provinces based on revenue, number of orders, average order value, and purchase frequency?**

This question matters because understanding differences in basic purchasing behavior helps Canada Vapes identify where demand is strongest and how customers engage with the brand in each region. By comparing these metrics, the company can determine whether certain provinces generate higher sales due to more customers, larger purchases, or more frequent buying. This insight is critical for deciding where to allocate marketing resources and how to tailor campaigns to match local buying patterns.

### **2. Which provinces have the highest-value customers based on repeat purchase behavior, average order value, and estimated customer lifetime value?**

This question is important because not all revenue is equally valuable. Provinces with loyal, repeat customers contribute more long-term value than those with mostly one-time buyers. Understanding where high-value customers are located allows Canada Vapes to focus retention strategies, loyalty programs, and personalized marketing efforts in regions that will generate sustained revenue growth.

### **3. What product categories are most preferred in each province, and how do these preferences vary by region?**

This question matters because product preferences can differ significantly across regions due to lifestyle, experience level, and regulatory factors. Identifying which products are most popular in each province helps Canada Vapes tailor its product offerings, promotions, and messaging. For example, some regions may prefer beginner-friendly kits, while others may favour advanced devices or refill

products. Aligning marketing with local preferences increases relevance and conversion rates.

4. **How do provincial regulations, vape taxes, and pricing impacts appear to influence customer purchasing behaviour?**

This question is critical because regulatory and pricing differences can directly affect demand. Higher taxes or stricter regulations may reduce purchase frequency, shift customers toward lower-cost products, or influence product selection. By understanding these impacts, Canada Vapes can adjust pricing strategies, product positioning, and promotional approaches to remain competitive while staying compliant with legal requirements.

5. **Which provinces appear to offer the strongest growth opportunity when revenue potential, customer value, regulatory difficulty, and competition are considered together?**

This question matters because growth decisions should not be based on a single factor such as current sales. A province with moderate sales but high customer value and manageable regulations may offer better long-term potential than a high-revenue but highly restricted market. This question helps Canada Vapes take a balanced, strategic approach to identifying priority regions for expansion and investment.

6. **Which underperforming provinces may represent missed opportunities rather than low-potential markets?**

This question is important because low sales do not always indicate low demand. Some provinces may underperform due to limited awareness, weaker marketing efforts, or logistical challenges. Identifying these hidden opportunities allows Canada Vapes to improve performance through targeted strategies rather than overlooking potentially profitable markets.

7. **What compliant marketing strategies should Canada Vapes use for each priority province?**

This question matters because insights are only valuable if they lead to actionable and compliant strategies. Given the strict regulations around vaping promotion, Canada Vapes must ensure that all marketing efforts are appropriate for legal-age consumers and follow advertising rules. This question ensures that the analysis translates into practical recommendations that drive growth while maintaining ethical and legal standards.

### 1.3 Connection to Concurrent Courses

This project connects to **Data Strategy and Decision Making** because it starts with a business problem and translates it into clear data questions, ensuring that the analysis supports informed decision-making (Provost & Fawcett, 2013).

It also connects to **Predictive Analytics** because it uses past customer behaviour to estimate future growth. For example, repeat purchases and spending patterns can help predict which provinces will experience stronger growth.

The project is also related to **Data Ethics and Governance** because vaping is a regulated product category. The analysis must respect privacy, comply with laws, and avoid targeting minors. All recommendations should be responsible and ethical (Government of Canada, 2020).

Finally, it connects to **Dashboard and Data Visualization** because the results need to be clearly presented. Charts, graphs, and dashboard summaries will help communicate findings effectively, track KPIs and support decision-making (Davenport & Harris, 2017).

## 2. Methodology & Methods Selection

### 2.1 Analytical Methodology

The project team will follow the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology to meet Canada Vapes' goal of finding the province's growth opportunities. CRISP-DM is a well-known framework with an iterative and structured way to address a business problem using data. It was chosen because it closely fits with the project's objective of converting data from transactions and markets into business recommendations, which involves business understanding, data preparation, data analysis, data evaluation, and reporting.

The project will combine **descriptive, diagnostic, and comparative analytical approaches** to evaluate customer purchasing behavior across Canadian provinces. Descriptive analytics will summarize historical sales performance and customer purchasing patterns, while diagnostic analytics will investigate the factors contributing to regional differences. Comparative analysis will be used to evaluate provinces against consistent business criteria, enabling the team to identify regions with the strongest growth potential. The findings from these analyses will support evidence-based recommendations that align with Canada Vapes' marketing and expansion objectives.

This analytical approach follows the 4 phases of the project with analysis of internal sales performance, external market research, identification of high-potential provinces, and the formulation of province-specific marketing recommendations.

## **2.2 Analytical Methods**

### **Data Preparation**

The project will begin by preparing the transactional dataset to ensure its quality, consistency, and reliability. The data will be reviewed for missing values, duplicate records, inconsistent province names, incorrect product classifications, and formatting issues. Additional business metrics, such as Average Order Value (AOV), purchase frequency, and repeat purchase rate, will be calculated where appropriate to support provincial comparisons.

### **Descriptive Analytics:**

Descriptive analytics will be applied to gain a general understanding of customers' buying behaviors in the Canadian provinces. Key performance indicators (KPIs) will be derived to show the overall performance of the business in the past and forecast trends in the region.

The planned analysis will involve:

- Total revenue by province
- Number of customers
- Number of transactions
- Average Order Value (AOV)
- Purchase frequency
- Top-selling product categories
- Repeat purchase behaviour

These metrics will directly contribute to the project's goal of understanding the difference in buyer behaviour and highlight the provinces that are currently doing well or may be worthy of further investigation.

### **Exploratory Data Analysis (EDA):**

Exploratory Data Analysis (EDA) will be conducted to understand the structure of the dataset, identify outliers, and examine relationships between customer behaviour, product preferences, and provincial performance. Visualizations such as bar charts, line charts, histograms, boxplots, scatter plots, and heatmaps will be used to identify patterns and communicate findings effectively.

**Comparative and Statistical Analysis:**

Comparative statistical techniques will be selected based on the characteristics of the dataset and the research questions. Where appropriate, the analysis may include correlation analysis, independent sample t-tests, Analysis of Variance (ANOVA), or Chi-square tests to determine whether differences in customer behavior across provinces are statistically significant. These methods will provide quantitative evidence to support business recommendations.

**Market and Regulatory Analysis:**

To complement the internal sales data, the project will include research on provincial vape regulations, taxation, advertising restrictions, retail competition, and market maturity. Information will be collected from reliable public sources such as provincial government websites, regulatory agencies, and industry reports. Comparing external market conditions with transactional data will help explain regional differences in customer behavior and identify factors influencing growth opportunities.

**Opportunity Assessment:** Each province will be evaluated using a consistent set of business criteria, including historical revenue, Average Order Value (AOV), purchase frequency, repeat purchase potential, customer value, market accessibility, competition, and provincial regulations. The combined assessment will enable the team to rank provinces and identify the two to three regions that offer the strongest growth potential.

**Strategic Recommendation Development:** Based on the analytical findings, the team will develop province-specific recommendations covering target customer characteristics, product focus, marketing messages, promotional strategies, and customer acquisition opportunities. These recommendations will support Canada Vapes in allocating marketing resources effectively while adapting strategies to regional market conditions.

**Data Sources**

The primary source of information will be the transactional dataset provided by Canada Vapes through the Riipen project. The dataset is expected to include customer purchase history, product categories, transaction values, and provincial sales information.



To support the analysis, secondary information will be collected from publicly available Canadian sources, including provincial government websites, regulatory agencies, taxation authorities, and industry publications. These sources will provide context regarding provincial regulations, vape taxes, market maturity, and competitive conditions, allowing internal sales data to be interpreted within the broader market environment.

3. Tools and Technologies

| Tools                     | Technologies                                                                                                                                                           |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Python (Jupyter Notebook) | Data cleaning, preprocessing, exploratory data analysis, statistical analysis, and trend analysis using libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn. |
| SQL                       | Extracting, filtering, joining, and summarizing transactional data for analysis.                                                                                       |
| Power BI                  | Developing interactive dashboards and visualizations to compare provincial performance and present findings to stakeholders.                                           |
| Microsoft Excel           | Initial data validation, quality checks, and preliminary calculations.                                                                                                 |
| WhatsApp and Riipen       | Team communication, collaboration, and document sharing throughout the project.                                                                                        |
| Microsoft PowerPoint      | Preparing the final stakeholder presentation and communicating project findings.                                                                                       |

4. High Level Project Timeline

| Project Phase                      | Timeline   |
|------------------------------------|------------|
| Project Planning                   | Week 8     |
| Data Collection & Preparation      | Week 8-9   |
| Exploratory & Descriptive Analysis | Week 9-10  |
| Market & Regulatory Research       | Week 10-11 |
| Comparative & Statistical Analysis | Week 11-12 |
| Opportunity Assessment             | Week 12-13 |
| Visualization & Report Development | Week 13-14 |
| Final Review & Presentation        | Week 15    |

### 5.1 Attached Canada Vapes Project Gantt Chart

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## 5.2 Attached Team Roles and Responsibility Matrix

| Role                                     |                                                                                                                                                                                                                                                                                                                                  |                    |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Role                                     | Responsibilities                                                                                                                                                                                                                                                                                                                 | Name               |
| <b>Project Manager / Team Lead</b>       | Responsible for project planning and execution, co-ordinating team activities, creating and tracking the project schedule, facilitating project meetings, liaising with the Industry Contact, risk management, completion of milestones, co-ordinating client reviews and final submission of project report.                    | Busayo Adebayo     |
| <b>Data Lead</b>                         | Gathers project data, evaluates data quality, cleans and preprocesses data, documents data metadata and data dictionary, performs exploratory data analysis, analyses the behaviour of buyers and purchasing trends by province and contributes to data-driven insights during project.                                          | Gift Oyugi         |
| <b>Research Lead</b>                     | Carries out research on provincial regulations, vape taxes, market conditions, competition and other external factors affecting buyer behaviour. Summarizes research findings to support the opportunity assessment and strategic recommendations.                                                                               | Damian Okelue      |
| <b>Strategy &amp; Marketing Lead</b>     | Conducts province evaluation criteria, ranks provinces based on growth opportunity, identifies priority provinces, and develops province specific marketing recommendations, target audience insights, messaging, promotional strategies, and product focus recommendations.                                                     | Abasiaka Anwanakak |
| <b>Visualization &amp; QA Lead</b>       | Assesses data quality, develops the Power BI dashboard prototype, creates charts and visualizations, supports dashboard development, coordinates quality assurance reviews, assists with report formatting, and develops presentation materials.                                                                                 | Sahil Kamane       |
| <b>Entire Team</b>                       | Collaborates throughout the project by participating in planning, reviewing findings, validating analyses, selecting priority provinces, reviewing marketing strategies, drafting the final report, reviewing project deliverables, finalizing submissions, delivering the final presentation, and documenting project feedback. | ALL                |
| <b>Industry Contact (Client Sponsor)</b> | Provides business objectives, clarifies the business context and supplies internal data, answers questions throughout the project, attends milestone reviews, responds to the findings, dashboards and recommendations, evaluates the final report and presentation.                                                             | Lauren Dietrich    |

## 6. Risk Assessment and Mitigation Strategies

To help ensure the successful delivery of this project, potential risks that could affect its execution were identified and assessed during the planning stage. These risks were based on the project's scope, expected analytical activities, stakeholder interactions, and the tools that will be used throughout the project. Appropriate mitigation strategies have also been proposed to reduce their potential impact and improve the likelihood of delivering accurate, high-quality results within the project timeline.

| <b>Risk</b>                                      | <b>Cause-Risk-Effect Statement</b>                                                                               | <b>Probability</b> | <b>Impact</b> | <b>Mitigation Strategy</b>                                                                                        |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------|---------------|-------------------------------------------------------------------------------------------------------------------|
| Inconsistent or Poor-Quality Historical Data     | Poor-quality historical data may reduce the accuracy of analysis and project recommendations.                    | Medium             | High          | Validate, clean, and standardize the data before analysis.                                                        |
| Scope Creep                                      | Additional project requests may expand the project's scope, resulting in increased workload and timeline delays. | Medium             | High          | Clearly define the project scope, document changes, and assess their impact before implementation.                |
| Analytical Bias and Misinterpretation of Results | Bias or incorrect interpretation of data may result in inaccurate findings and recommendations.                  | Medium             | High          | Review findings as a team and ensure conclusions are supported by evidence from the data.                         |
| Technical Issues with Tools and Software         | Technical issues with Power BI, Python, SQL, or Excel may interrupt project activities and delay analysis.       | Low                | Medium        | Regularly back up project files, test tools in advance, and have alternative solutions available if issues occur. |
| Project Time Constraints                         | Limited project time may reduce data validation and quality checks, affecting                                    | High               | High          | Set clear milestones, prioritize critical tasks, and allocate enough time for data                                |

|                                               |                                                                                                                      |        |        |                                                                                                            |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|--------|------------------------------------------------------------------------------------------------------------|
|                                               | the accuracy of project results.                                                                                     |        |        | validation and final review.                                                                               |
| Misinterpretation of Stakeholder Requirements | Misunderstanding stakeholder requirements may result in analysis that does not fully address the project objectives. | Medium | High   | Maintain regular communication with stakeholders and confirm requirements before major project activities. |
| Communication Delays with Stakeholders        | Delays in stakeholder communication may slow decision-making and affect project progress.                            | Medium | Medium | Schedule regular check-ins and communicate updates promptly.                                               |

## 7. Data governance and ethics checklist

| <b>Data governance/ethics</b> | <b>Application</b>                                                                                            |
|-------------------------------|---------------------------------------------------------------------------------------------------------------|
| PIPEDA Principles             | Any personal information included in the project dataset will be handled in accordance with PIPEDA principles |
| Purpose Limitation            | Project data will only be used to answer the agreed research questions for Canada Vapes.                      |
| Data Minimization             | Only data necessary to achieve the project objectives were requested and will be analyzed.                    |
| Confidentiality               | Access to project data will be limited to authorized team members throughout the project.                     |
| Transparency                  | Analytical methods, assumptions, and limitations will be clearly documented in the final report.              |
| Accountability                | Team members will maintain clear documentation of project decisions and analytical processes.                 |
| Responsible Data Use          | All project data will be used ethically and only for the intended purpose.                                    |

## 8. References

Canada Revenue Agency. (2026). *Additional specified vaping province effective February 26, 2026*. <https://www.canada.ca>

Canada Vapes. (n.d.). *About Canada Vapes*. <https://canadavapes.com>

Davenport, T. H., & Harris, J. G. (2017). *Competing on analytics: The new science of winning* (Updated ed.). Harvard Business Review Press. <https://hbr.org/product/competing-on-analytics-the-new-science-of-winning-updated-edition/10070-HBK-ENG>

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